

Smart Grid Digital Twin Dashboard - Client Overview

Enterprise Smart Grid platform based on Vue.js, Quasar, EMQX MQTT, CrateDB, AI Analytics, GIS and Digital Twin technologies.

Executive Dashboard

Realtime KPI monitoring, grid health, alarms, load and operational awareness.

Grid Map

Interactive GIS map with substations, transformers, power lines, customers and regions.

Digital Twin

Region, substation and transformer digital twins with realtime telemetry.

Alarm Center

Alarm management, root cause analysis and event correlation.

AI Forecasting

Load forecasting, blackout prediction and predictive insights.

Predictive Maintenance

RUL models, failure probability and spare parts forecasting.

Weather Intelligence

Storm tracking, lightning risk and weather overlays.

Risk & Heatmap Layers

AI risk scoring and operational heatmaps across the network.

Realtime Analytics

CrateDB-powered realtime and historical analytics.

System Topology

Visualization of SCADA, EMQX, CrateDB, AI and Digital Twin services.

Key Enterprise Features

- Realtime MQTT → EMQX → WebSocket architecture
- CrateDB realtime analytics and KPI aggregation
- Mapbox GIS and Deck.gl power line rendering
- AI Load Forecasting and Blackout Prediction
- Alarm Correlation and Root Cause Analysis
- Weather Risk Intelligence
- Digital Twin drill-down for Regions, Substations and Transformers
- Predictive Maintenance and Remaining Useful Life models
- Enterprise monitoring and system topology visualization